

Breast Ultrasound Lesion Segmentation: Research Write-up

Version 1.0.0 · Site-only research report (not a journal submission)

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Live demo: <https://surabhif.github.io/Breast-ultrasound-lesion-segmentation/>

Repository: <https://github.com/surabhif/Breast-ultrasound-lesion-segmentation>

Software DOI: <https://doi.org/10.5281/zenodo.23286597>

Research demo — not for clinical use. Not for diagnosis, screening, or care decisions.

1. Abstract

Breast ultrasound helps doctors look at breast lumps, especially when mammograms are hard to read in dense tissue. This student project builds a program that outlines those lumps on ultrasound pictures and estimates how well it does at telling harmless from cancerous lumps. It was trained on one public image collection and checked on three others from different hospitals. The main quality measure is an outline-overlap score (Dice), which is high when the computer outline matches the expert outline. On images saved for testing, the browser version reaches 0.697 overall. On images that contain a lump, the outline match is 0.764. On two outside collections, outline scores stay similar (0.714 and 0.627), while telling harmless from cancerous lumps gets harder after the hospital and scanner change. On a third outside collection, pictures without lumps remain difficult: false outlines pull the overall score down to 0.386, even though on images that contain a lump the outline match is 0.679. The write-up also studies near-identical photos, burned-in measurement marks, and whether outlines stay stable when a picture is flipped. A retrained version did better on some outside images but not consistently, so the original stays. This is an educational research demonstration, not a medical device, and must not be used for diagnosis or care decisions.

Key findings in plain English

- On images saved for testing, the browser program's outline-overlap score is 0.697 overall. On images that contain a lump, the outline match is 0.764.
- On outside hospitals, outline quality stays in a similar ballpark on BUS-BRA (0.714) and BrEaST (0.627), but telling harmless from cancerous lumps gets harder (0.638 / 0.721 vs 0.931 on the original test set).
- On BUS-UCLM, pictures without lumps often get false outlines (320/413). On images that contain a lump, the outline match is 0.679 — the fairer comparison.
- Near-identical photos are common (322 images sit in multi-member groups). Checking for near-identical photos showed they did not inflate the scores, but patient IDs are still missing from the main training set.
- Erasing measurement marks and retraining helped on some outside images, but not consistently enough to replace the published model, so the original stays (version 1.0.0).
- This is an educational research demo only — not for diagnosis, screening, or care decisions.

2. Background: why outlining lumps on ultrasound matters

Breast ultrasound uses high-frequency sound waves to form a live picture of breast tissue. Clinicians often add ultrasound after a mammogram or a physical exam when they need a closer look at a mass, especially in dense breasts where X-ray contrast is limited (Mendelson et al., ACR BI-RADS Ultrasound, 2013). For a patient, the practical questions are simple: Is there a lump? How large is it? Does it look more concerning or more reassuring? Those answers influence biopsy decisions and, later, surgical planning.

Outlining a lump — drawing its boundary on the image — is the spatial version of that conversation. A careful outline supports size estimates, shape descriptors used in Breast Imaging Reporting and Data System (BI-RADS) language, and communication among radiologists, surgeons, and patients. Automated outlining is attractive for education and research because public datasets now include expert masks. It is also easy to over-claim: a computer outline is not a surgical margin, an imaging diameter is not pathologic tumor stage, and a model score is not a BI-RADS assessment category (Moran et al., SSO-ASTRO margin guideline, 2014).

This project asks a narrower, honest question suitable for a student portfolio: can a small model outline lesions in the browser, and do the published numbers survive outside the training hospital's pictures? The site pairs the demo with BI-RADS context and a surgeon's-view page so readers see what the numbers do not mean.

From a patient's point of view, an outline is a picture of extent: how much of the frame looks abnormal, and whether the edge looks smooth or irregular. Surgeons later care about margins in tissue, not pixels, which is why this write-up repeatedly separates imaging size from pathologic stage. From a student's point of view, the same outline is a measurable prediction that can be scored, stress-tested on outside hospitals, and shipped in a browser so anyone can see the failure modes without installing a machine-learning stack.

3. Related work

U-Net-style encoder-decoder networks remain a default for biomedical segmentation (Ronneberger et al., 2015). On the Breast Ultrasound Images (BUSI) dataset, recent papers report a wide range of scores depending on how train and test pictures are chosen, whether near-identical photos are removed, and whether pictures without lumps are included. UNeXt (Valanarasu & Patel, MICCAI 2022) reports F1 about 0.79 and intersection-over-union (IoU) about 0.67 on an 80-20 mix of harmless and cancerous BUSI frames at 256×256 — a useful context number, but not an apples-to-apples match for this project's near-identical photo grouping and per-image outline-overlap score. Musah et al. (2025) show that BUSI scores fall after de-duplication and fall further when a BUSI-trained model is tested on BrEaST (reported Dice about 0.49 in their setting). Pawłowska et al. (2023) document roughly 235 duplicated BUSI images (~19%), which is why this project groups near-identical photos before splitting.

This work's browser model reaches outline-overlap score 0.697 (on images that contain a lump, 0.764) on its own test set, and outside outline-overlap scores 0.714 (BUS-BRA), 0.627 (BrEaST), and 0.679 on BUS-UCLM images that contain a lump. Area under the curve for telling harmless from cancerous lumps drops from 0.931 at home to 0.638 / 0.721 / 0.780 outside — the same direction Wang (2026) reports under dataset shift. Published tables are context, not a leaderboard; method differences dominate small score gaps. Full row-by-row notes live in the project's literature-comparison document.

Positioning matters as much as the table. This is not a claim to beat UNeXt or nnU-Net on a public leaderboard. It is a claim that a single frozen browser artifact has internal scores, external scores, cleaning experiments, and a written rule for when the artifact may change — and that those pieces are visible next to the demo.

Paper / setting	Split & notes	Reported figure	How it compares here
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This work v1.0.0 served	BUSI own test n=112	Outline-overlap 0.697; with lump 0.764; AUC 0.931	Reference row
This work outside hospitals	Frozen weights; no retune	BUS-BRA 0.714; BrEaST 0.627; BUS-UCLM with lump 0.679	Transfer check
Valanarasu & Patel, UNeXt (2022)	BUSI B+M; 80-20 mix; 256 ²	F1 $\approx$ 0.79; IoU $\approx$ 0.67	Partly comparable (different split and aggregation)
Musah et al. (2025)	BUSI→BrEaST OOD among other settings	BUSI→BrEaST Dice $\approx$ 0.49 in their setting	Same OOD direction; different model/resolution
Pawłowska et al. (2023)	BUSI audit	~235 duplicates (~19%)	Motivation for near-duplicate grouping

Takeaway: published BUSI scores are easy to misread unless the split and metric definition match; this project optimizes for honest transfer reporting, not a leaderboard win.

4. Data

Four public breast-ultrasound collections are used. BUSI is the training and internal test source. BUS-BRA, BrEaST, and BUS-UCLM are external checks of the frozen served model with no threshold retuning. BUSI provides no patient identifiers; the other three do. Annotation marks (calipers, text overlays) were flagged heuristically on 46.9% of BUSI frames (366 / 780). Perceptual near-duplicate grouping produced 601 groups, with 322 images in multi-member groups.

Dataset	Images	Patients	Source country	Scanner notes	License	Excluded / notes
BUSI (Al-Dhabyani et al., 2020)	780	Not published	Egypt (Baheya / Cairo University collection)	Clinical ultrasound archive; device mix not standardized in the release	Citation required; full-archive redistribution rights unclear — not redistributed here	Multi-mask lesions OR-merged; near-duplicate groups kept intact across splits
BUS-BRA (Gómez-Flores et al., 2024)	1875	1064	Brazil	GE Logiq 5/7, Toshiba Aplio 300, U-Systems (device field in release)	Creative Commons Attribution 4.0 (Zenodo)	No normal class in this release; used as external test of frozen weights
BrEaST (Pawłowska et al., 2024)	256	256	Poland	Clinical scanners with physical pixel size in metadata	Creative Commons Attribution 4.0 (TCIA)	Freehand tumour masks; few normals; used as external test
BUS-UCLM (Vallez et al., 2025)	640	38	Spain (Castilla-La Mancha)	Siemens ACUSON S2000	Creative Commons Attribution 4.0 (Mendeley)	43 Doppler/combined frames excluded; images scored locally, not redistributed

Takeaway: four public collections from different countries and scanners; only BUSI trains the published model, and BUS-UCLM drops Doppler/combined frames before scoring.

Sources: dataset papers cited above; counts from committed results JSON (cleaning_experiment.json; external busbra/breast/busuclm.json).

BUS-UCLM (Vallez et al., 2025; Mendeley Creative Commons Attribution 4.0) is scored for the frozen browser model: $n=640$ / 38 patients (43 Doppler/combined frames excluded). Overall outline-overlap score 0.386 [0.326, 0.449]; on images that contain a lump, 0.679 [0.593, 0.755]; area under the receiver-operating curve (AUC) 0.780. Normal false positives dominate the overall figure (320/413) — the same known weakness as on BUSI (12/19). Prefer the score on images that contain a lump when comparing across collections (between BrEaST and BUS-BRA).

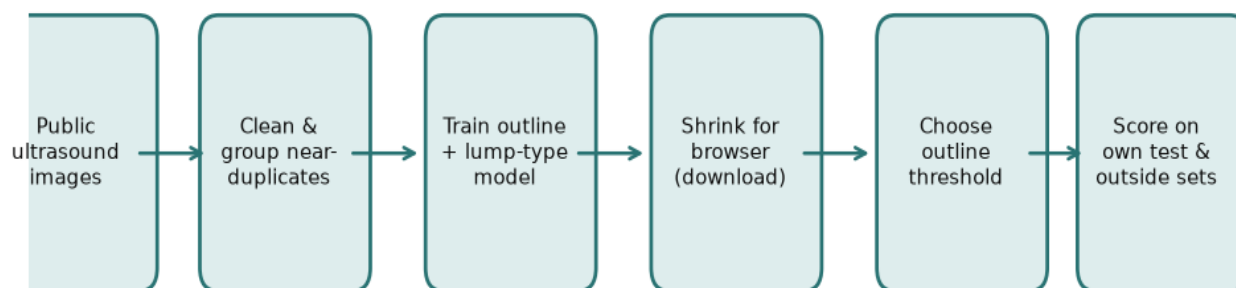
Split sizes after keeping near-identical photo groups together: train 534, validation 134, test 112.

5. Methods

5.1 Pipeline overview

Figure 1 summarizes the path from public images to browser scores. Every served number on the site is produced by regenerating metrics from committed results files so the demo and the write-up cannot silently diverge.

Figure 1. End-to-end pipeline (plain overview)



Takeaway: images are cleaned and grouped, a compact model is trained, shrunk for browser download, thresholded, then scored on the project's own test images and on outside hospitals.

5.2 Preprocessing

Each image is loaded as red-green-blue, resized to 160×160 with a shared half-pixel-center bilinear resize (identical in Python evaluation and in the browser), scaled to [0, 1], and normalized with ImageNet channel means and standard deviations. Expert masks are resized with nearest-neighbour interpolation and binarized; when a lesion has several mask files they are combined with a logical OR so any marked region counts.

5.3 Near-duplicate grouping

Because BUSI does not publish patient identifiers, near-identical frames can otherwise appear in both training and testing. Frames are grouped with a perceptual hash so that an entire near-duplicate group stays on one side of each split. That does not prove zero leakage — only that the obvious near-copies are not split apart.

5.4 Model, explained intuitively

The network is a compact encoder-decoder in the U-Net family (Ronneberger et al., 2015) with a ResNet-18 ImageNet encoder (He et al., 2016). Intuitively, the encoder compresses the ultrasound into a smaller grid of features; the decoder expands those features back to a full-size outline map. A second head reads the same compressed features and outputs a single score for benign versus malignant. Training uses paired image-mask augmentations, stratified grouped splits, and checkpoint selection on lesion outline-overlap score on the validation set.

5.5 Shrinking the model for the browser

The full-precision training checkpoint is about 46.3 megabytes. For the website it is exported to the Open Neural Network Exchange (ONNX) format and dynamically quantized to 8-bit integer weights (often called INT8), yielding a 11.7 MB download. The browser runs that artifact with onnxruntime-web (WebAssembly) inside a Web Worker so the page stays responsive. Post-processing matches Python evaluation: probability threshold 0.4 and minimum connected-component area 40 pixels.

5.6 Threshold choice

The outline threshold 0.4 and minimum area 40 were chosen on BUSI validation lesion outline-overlap score and then frozen. External sets reuse the same operating point with no retuning — an intentional honesty choice that can look worse than a tuned baseline but answers the transfer question readers actually care about.

5.7 Training details in brief

Training uses the grouped BUSI split (train 534, validation 134, test 112). The optimization target emphasizes overlap with expert masks; the classification head is trained jointly so the same backbone supports both outlining and telling harmless from cancerous lumps. Checkpoints are picked by validation outline-overlap on images that contain a lump, not by hunting for the best final test number after the fact. The served fingerprint (sha256 prefix 0bbf529d...) is the artifact evaluated everywhere in this write-up unless a section explicitly names a later retrain experiment.

Evaluation code paths are shared: the same resize, normalization, threshold, and minimum-area filter run in Python metrics and in the browser worker. That parity is why the Results page can honestly say the site numbers are the served-model numbers, not a prettier training checkpoint left behind in a notebook.

6. Results

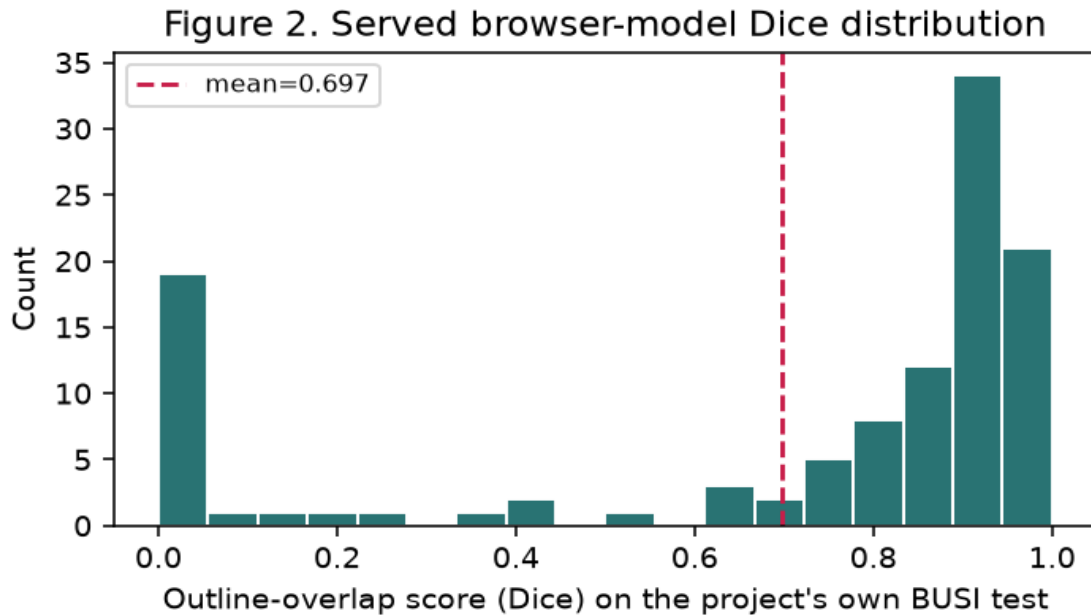
6.1 Results on the project's own test images

Table 2 compares the full-precision training checkpoint with the browser version on the same test images (n=112), kept apart from training by near-identical photo grouping. Overall outline-overlap scores are close (0.686 vs 0.697). On images that contain a lump, the scores are 0.751 vs 0.764. Per-class browser scores: harmless lumps 0.749 (n=63), cancerous lumps 0.797 (n=30). Pictures without lumps remain hard: non-empty predicted outlines on 12/19 normals.

Metric	Full-precision training	Served browser model
Outline-overlap score (all)	0.686	0.697
Outline-overlap (images with a lump)	0.751	0.764
Intersection-over-union (IoU)	0.618	0.630
AUC (telling harmless from cancerous)	0.939	0.931
Sensitivity / Specificity	0.867 / 0.873	0.867 / 0.841
Expected calibration error (ECE)	0.111	0.111
False outlines on normals	12/19	12/19

Takeaway: shrinking the model for the browser barely changes outline quality on the project's own test images; empty/normal pictures are still the weak spot.

Sources: results/full_run.json, results/served_int8_test.json.



Takeaway: most of the project's own test cases land at high outline-overlap scores, with a long left tail of hard misses and false outlines on normals.

6.2 Results on outside hospitals

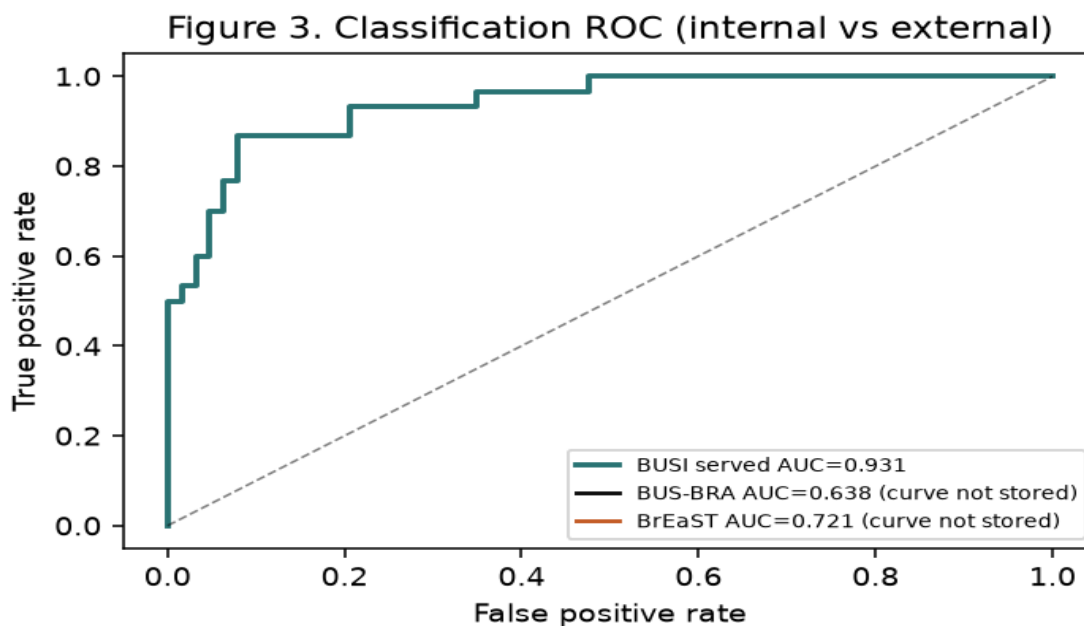
The same frozen browser model was scored without retuning on BUS-BRA (n=1875, 1064 patients), BrEaST (n=256, 256 patients), and BUS-UCLM (n=640, 38 patients). Per-class outline-overlap scores: BUS-BRA harmless 0.710 / cancerous 0.721; BrEaST harmless 0.593 / cancerous 0.685; BUS-UCLM

harmless 0.702 / cancerous 0.628.

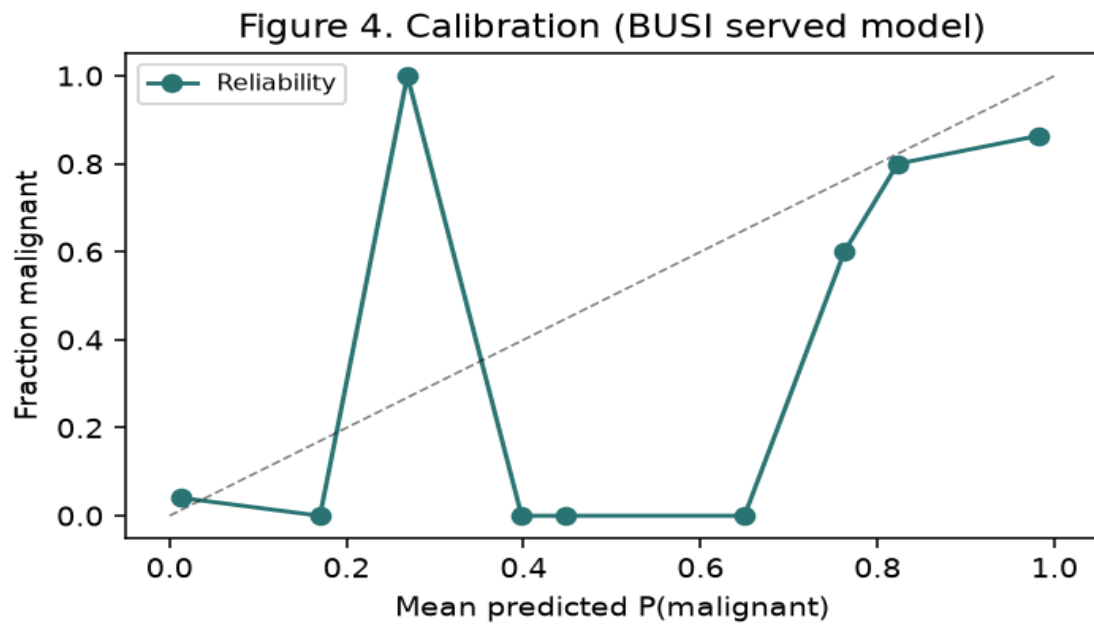
Set	Outline-overlap	Images with a lump	AUC
BUSI (own test)	0.697	0.764	0.931
BUS-BRA	0.714	0.714	0.638
BrEaST	0.627	0.629	0.721
BUS-UCLM	0.386	0.679	0.780

Takeaway: outlines travel better than telling harmless from cancerous lumps; BUS-UCLM's overall figure is dragged down by false outlines on normals (320/413).

Sources: results/external/busbra.json, breast.json, busuclm.json.



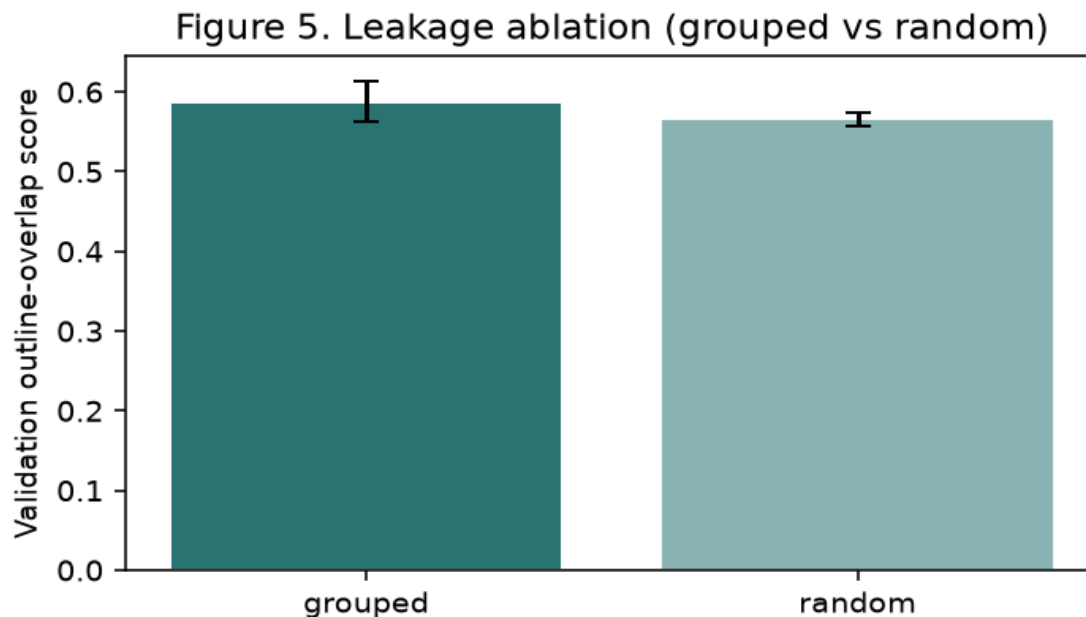
Takeaway: the curve that separates benign from malignant looks strong on BUSI and weaker once the hospital and scanner change.



Takeaway: predicted cancer probabilities are only roughly aligned with observed rates (ECE 0.111) — confidence should be read cautiously.

6.3 Checking for near-identical photos

A short training schedule (6 epochs × seeds 42,43,44) compared keeping near-identical photos together versus mixing them across train and check sets, then scored both on the same careful test set. Checking for near-identical photos showed they did not inflate the scores (difference about -0.022). Risk remains because patient identifiers are absent and near-copies are documented in the literature.



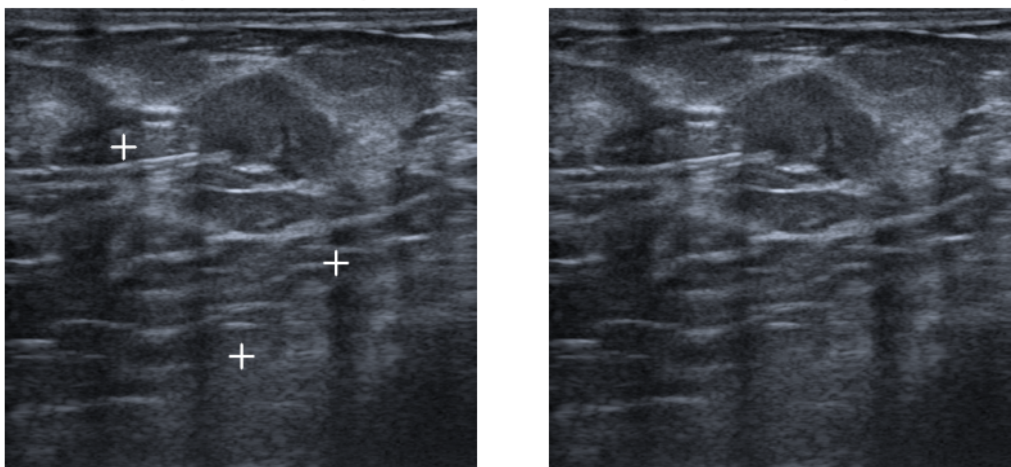
Takeaway: on this short check, mixing near-identical photos did not inflate check-set scores — but missing patient IDs are still a reason for caution.

Source: results/leakage_ablation.json.

6.4 Measurement-mark experiments

Images with burned-in measurement marks often look easier: marked-frame outline-overlap score 0.764 versus unmarked 0.628. Erasing markers on the original test moved overall score from 0.697 to 0.702 and did not close that gap. Retraining with erased marks produced outside scores BUS-BRA 0.626 and BrEaST 0.608, with unmarked-image score 0.626 and AUC 0.945 — not enough to replace served v1.0.0 under the written replacement criteria.

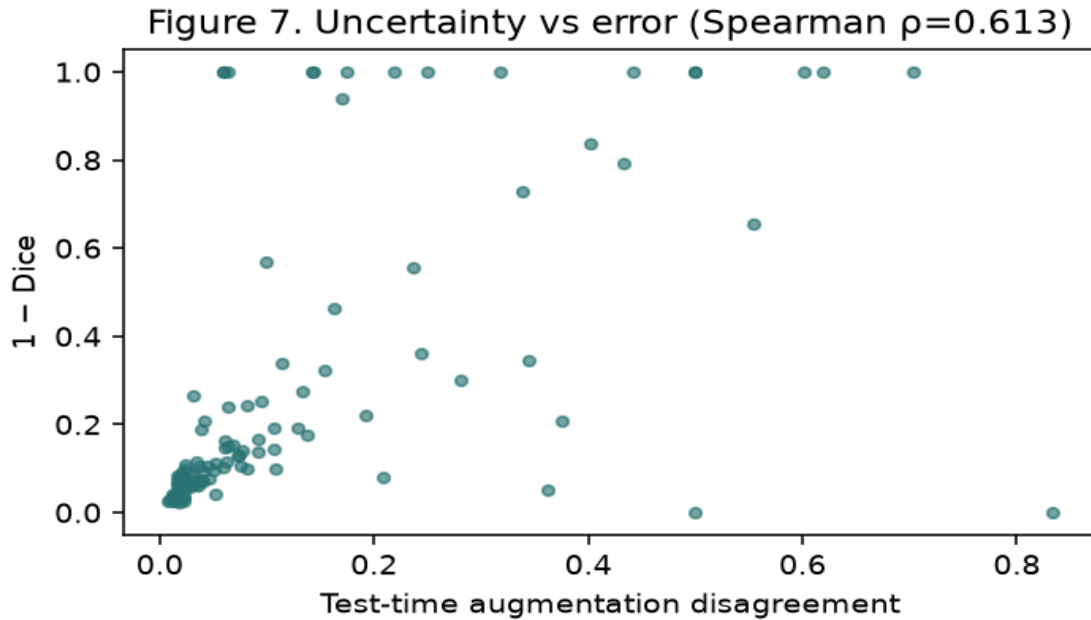
Figure 6. Caliper erase demo (CC BY BrEaST; digitally altered)
Before (synthetic calipers) After Telea inpaint



Takeaway: measurement marks can act as shortcuts; simply erasing them on test images does not automatically make hard unmarked cases easy.

6.5 When the outline stays stable under flips

Test-time augmentation (TTA) flips and mild transforms are run offline; disagreement across those views is treated as an uncertainty signal. Spearman correlation between disagreement and 1–Dice is 0.613 on the BUSI test (n=112). Keeping the most certain 80% of cases raises mean outline-overlap score to 0.778. Interpretation: agreement under flips — not a calibrated probability of being wrong.



Takeaway: when the outline stays stable under flips, errors are less common — useful as a research triage signal, not as a clinical confidence meter.

6.6 Size agreement with experts (BrEaST)

On BrEaST, model versus expert longest diameter (mm) shows Pearson correlation proxy 0.697 (n=252); area correlation proxy 0.763; disagreement at a 20 mm threshold occurs on about 17.1% of cases. Research only — imaging size is not pathologic T-stage.

6.7 A later retrain, and why the original stayed

A later candidate trained on BUSI plus BUS-BRA (larger backbone, 256×256, stronger augmentation) was compared with served v1.0.0 under written replacement criteria set beforehand. Average outline-overlap on unmarked test images was 0.614 versus v1 0.623 (difference -0.009). BUS-BRA images from the same source rose to 0.894, and truly outside BrEaST to 0.672, with AUC 0.912 and packaged size about 21.3 MB. A retrained version did better on some outside images but not consistently, so the original stays (v1.0.0); the comparison is published as an experiment.

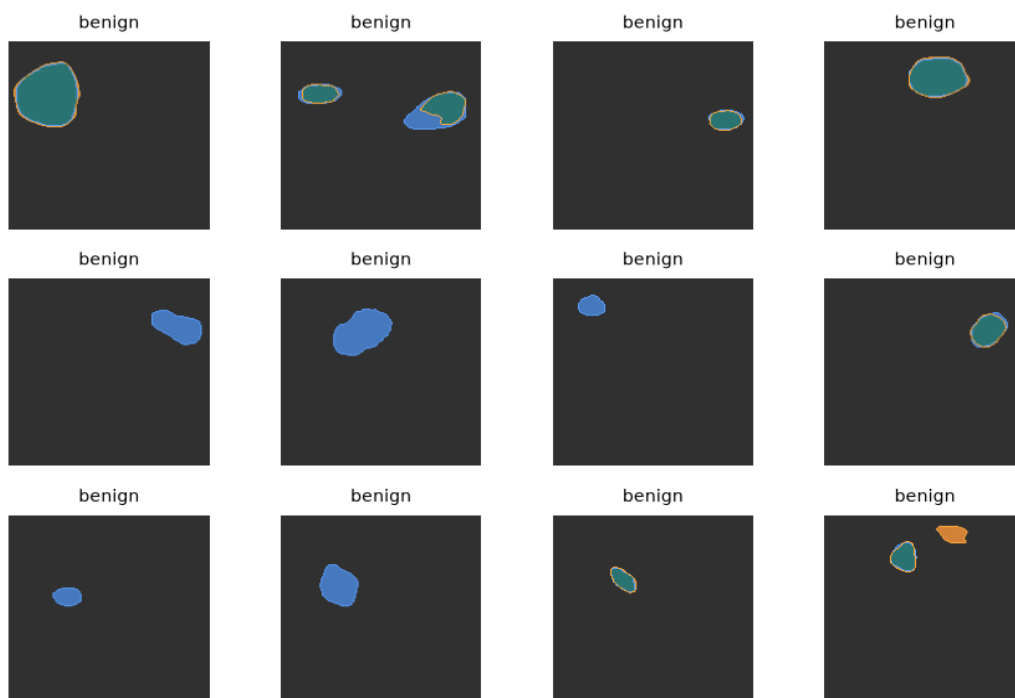
Takeaway: a flashier chart on one check set is not enough; the published model only changes when every pre-agreed gate passes on the average across training seeds.

Source: results/v2_experiment.json.

7. Where the model goes wrong

The Model Errors explorer on the site lists 105 hard test cases from the project's own collection as outline-only silhouettes (BUSI ultrasound pixels withheld because redistribution rights are unclear). Category counts: boundary disagreement 65, false lump on a normal picture 12, wrong class 11, missed lump 8, over-segmentation 7, under-segmentation 2. Boundary disagreements dominate; false outlines on normals remain the most intuitive failure mode for a triage-style demo.

Figure 8. Failure-case outline silhouettes (BUSI; no ultrasound pixels)



Takeaway: silhouette cards make misses visible without redistributing BUSI pixels — most hard cases are boundary fights, not total collapses.

8. Discussion: what the numbers mean in practice

An outline-overlap score near 0.697 on the project's own test images means that, on average, the computer outline and the expert outline share most of their area — useful for a demo and for learning what these scores feel like, not for claiming radiologist parity. Outside outline scores in the 0.63–0.71 range suggest drawing the boundary travels better than telling harmless from cancerous lumps, which falls when scanners and labeling habits change. That pattern matches the broader literature on dataset shift.

Normal false positives (12/19 on BUSI; 320/413 on BUS-UCLM) are the practical warning label: a browser demo that “finds” a lump on a healthy-looking frame can mislead a casual reader faster than a slightly jagged outline on a true mass. Calibration (ECE 0.111) is imperfect, so verbal confidence cues are intentionally subdued on the site.

Readers comparing this project with published BUSI leaderboards should weight method first. Many papers mix near-identical photos across train and test; this project keeps those groups together, reports the browser download (11.7 MB), and publishes outside-hospital numbers without retuning. A higher home-set score under a leakier setup is not a stronger scientific claim than a slightly lower score under a stricter one.

Keeping the original model is part of the result, not something to hide. BUS-BRA scores on the same-source check set rose to about 0.894, and BrEaST to about 0.672, yet the unmarked-image average (0.614 vs v1 0.623) did not clear the bar written beforehand. Setting that bar before seeing the average is how a student project stays honest when a larger model looks tempting on a chart.

For admissions readers and scientifically literate parents, the portfolio claim is process honesty: careful train/test grouping, frozen outside tests, written replacement criteria that refused a flashy upgrade, and a downloadable write-up whose figures are locked to committed JSON.

9. Limitations

Several constraints bound how far these numbers should travel outside a research demo:

- No patient identifiers on BUSI → residual risk from near-identical photos despite perceptual grouping.
- Domain shift: classification AUC drops from 0.931 internally to as low as 0.638 on BUS-BRA.
- Calipers and on-screen text may still act as shortcuts on annotated frames.
- 160×160 resolution loses fine boundary detail; normal-image false positives remain (12/19).
- Single-author student project without clinical validation; not a medical device.
- A model score is not a BI-RADS category; imaging size is not surgical staging.
- External BUS-UCLM images were scored locally and are not redistributed with the site.

10. Ethics and responsible use

Code is released under the MIT license. The full BUSI archive is not redistributed (redistribution rights are unclear); users must obtain it themselves and cite Al-Dhabyani et al. (2020). External demo imagery uses Creative Commons Attribution collections with credit. The site uses no analytics or cookies; Demo images are processed in the browser only and are not uploaded to a server. This write-up and the live demo are educational research materials. They must not be used for diagnosis, triage, screening, or any care decision. No clinician reviewed this project for clinical deployment, and none is claimed.

11. Future work

Natural next steps, if pursued, include stronger normal-vs-lesion rejection, patient-level collections with clearer licenses, higher-resolution models that still fit a browser budget, and uncertainty displays that a non-specialist can interpret without over-trust. Any promoted weight file would need a new version number and a fresh pass of the written replacement criteria.

A second thread of future work is communication: shorter silhouette cards that teach error types, clearer captions for parents who are not engineers, and printable one-pagers that keep the same locked numbers as this PDF. None of those changes should quietly retune thresholds on external data.

12. Reproducibility

Software version 1.0.0 is archived with DOI <https://doi.org/10.5281/zenodo.23286597>. To regenerate this PDF: install Python dependencies (reportlab, pypdf, matplotlib), run the report builder script from the repository root, then run the drift check so every locked number still appears in the PDF text. Paper metrics are exported from the same results JSON into LaTeX macros. Served artifact fingerprint (sha256 prefix): 0bbf529d... · packaged size 11.7 MB. Typical software stack for regeneration: Python 3.10+, Node.js 20+ for the web app, onnxruntime / onnxruntime-web for model execution. Exact pinned versions live in the repository lockfiles on the tagged release.

13. AI-use disclosure

Project by Surabhi Fadnavis. The code, analysis, text, and video were produced with AI tools (Cursor).

14. Glossary

Outline-overlap score (Dice)

A number from 0 to 1 describing how much the computer outline and the expert outline overlap. Twice the shared area divided by the sum of both areas; 1 is a perfect match.

Intersection-over-union (IoU)

Shared outline area divided by the area of either outline. Related to Dice; usually a little lower than Dice for the same pair of outlines.

Area under the curve (AUC)

A summary of how well a score ranks cancerous cases above benign ones across all thresholds. 0.5 is chance; 1.0 is perfect ranking.

Sensitivity / Specificity

Sensitivity is the share of truly cancerous cases the score catches at a chosen threshold; specificity is the share of truly benign cases it correctly leaves below that threshold.

Calibration / expected calibration error (ECE)

Whether a predicted probability (for example “70% malignant”) matches how often such cases are actually malignant. ECE summarizes the mismatch across probability bins.

BI-RADS

Breast Imaging Reporting and Data System — a standard lexicon and assessment categories radiologists use. A model score is not a BI-RADS category.

U-Net

A widely used neural-network shape for image outlining: a compressing path, an expanding path, and skip connections that preserve spatial detail.

ONNX

Open Neural Network Exchange — a portable file format for trained models so the same weights can run in Python and in the browser.

Quantization (INT8)

Storing weights with 8-bit integers instead of 32-bit floats to shrink download size and speed inference, with small metric changes when done carefully.

Test-time augmentation (TTA)

Running the model on flipped or lightly altered copies of the same image and measuring how much the outlines disagree — used here as an uncertainty signal.

External validation

Scoring a frozen model on data from other hospitals or collections without retuning, to test whether results were specific to the training source.

Data leakage

Accidental overlap between training and test information (for example near-duplicate frames on both sides) that makes scores look better than true generalization.

15. References

Citations below support the clinical framing, datasets, model family, and software archive. Prefer the dataset papers when quoting collection size or license.

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